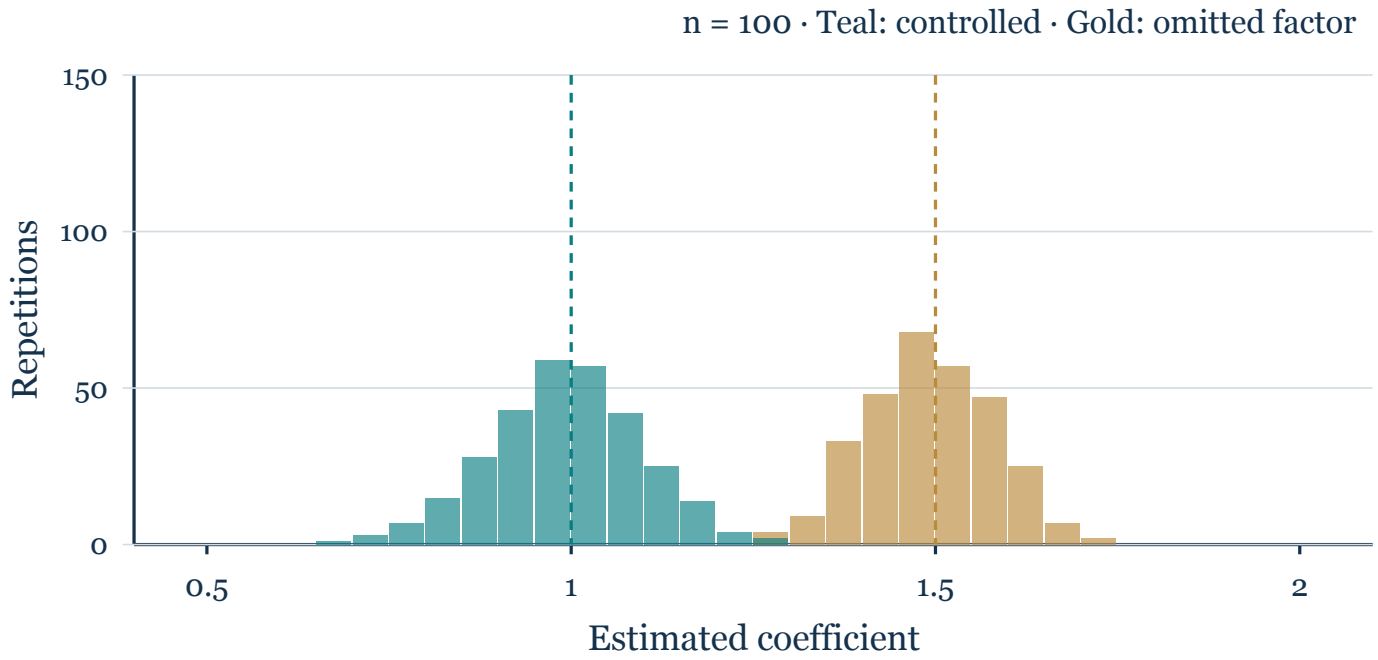
Can a larger dataset rescue a biased design?

Use a known population to separate bias from sampling noise.



Actual Stata Monte Carlo estimates: 300 repetitions at each sample size. The data-generating staffing coefficient is 1; the omitted-variable target is 1.5.

Define the simulation as an economic thought experiment

The generated truth is known by construction.

> $X = A + V$

> $Y = 1 + X + A + U$

> A, V, U independent, with mean zero and variance one

Unbiasedness and consistency ask different questions

One concerns an average at a sample size; the other concerns a limit.

> Unbiasedness: correct repeated-sample mean at a fixed sample size.

> Estimator consistency: concentration near the target as sample size grows.

> Causal consistency: observed/potential-outcome link; a different use of the word.

The law of large numbers stabilises averages

More data can reveal a stable but unwanted relationship.

> Law of large numbers: sample averages approach population means under stated conditions.

> Consistent estimator: concentrates near its target as the sample grows.

Measure co-movement before calculating the limit

Covariance records whether centered variables tend to move together.

> $\operatorname{Cov}(X, A) = E[(X - EX)(A - EA)]$

> $X = A + V: \quad \operatorname{Var}(X) = 2, \quad \operatorname{Cov}(X, A) = 1$

Calculate the wrong limit before simulating it

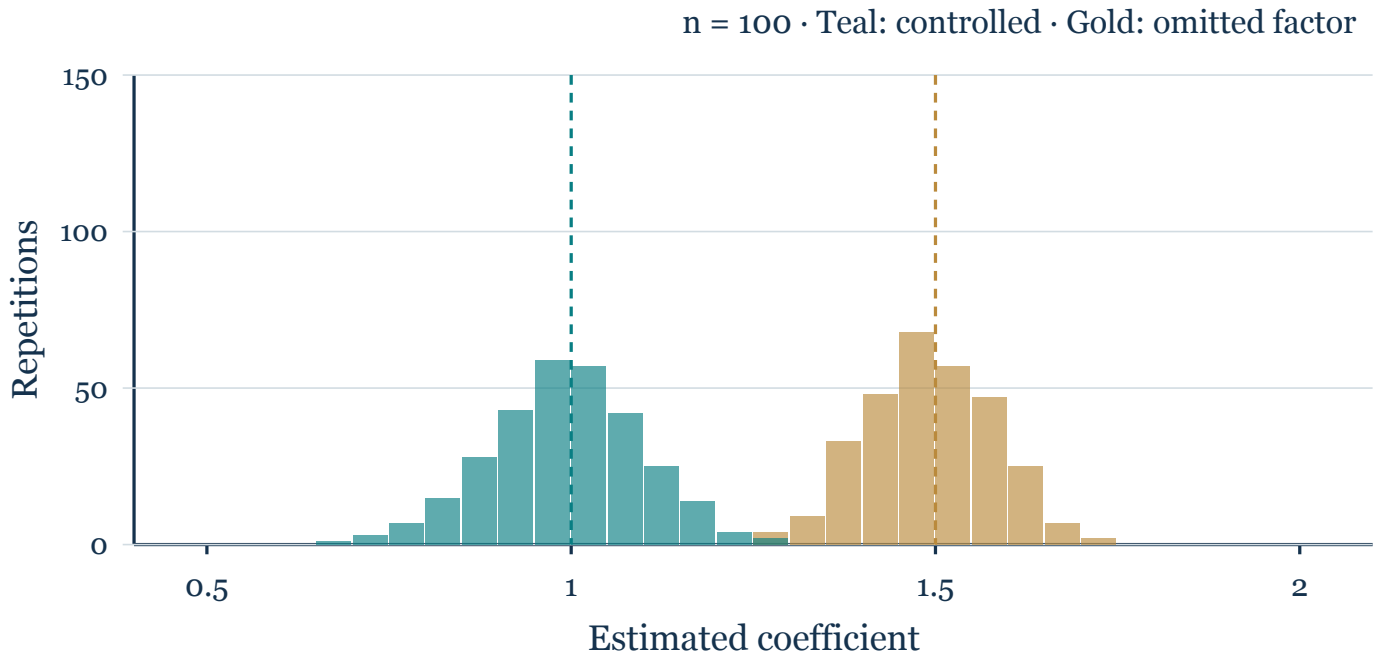
The short slope tends to one and a half.

$$\delta_1 = \frac{\operatorname{Cov}(X, A)}{\operatorname{Var}(X)} = \frac{1}{2}$$

$$\operatorname{plim} \widehat{b}_{\text{short}} = 1 + \frac{1}{2} = 1.5$$

Compare repeated samples at two sizes

Concentration alone is not evidence of a correct target.



Actual Stata Monte Carlo estimates: 300 repetitions at each sample size. The data-generating staffing coefficient is 1; the omitted-variable target is 1.5.

More data, same bias

Try the comparison before opening the explanation.

Practice

An estimator converges to 1.4 while the causal target is 1.0. What happens to the bias problem as n grows?

More data, same bias: explanation

Follow the calculation and interpret its units.

> It remains: the estimator is inconsistent for the causal target. Sampling variation may shrink around the wrong limit. Narrow intervals around 1.4 do not repair the design.

Read one large generated sample

A small SE can accompany substantial omitted-variable bias.

```
. regress y x, vce(robust)

------------------------------------------------------------------------------
             |               Robust
           y | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
           x |   1.516317   .0139733   108.51   0.000     1.488921    1.543712
       _cons |   .9947365   .0194616    51.11   0.000     .9565808    1.032892
------------------------------------------------------------------------------
```

The central limit theorem describes scaled fluctuations

The square-root rate concerns sampling variation around a limit.

> Average of independent zero-or-one draws, each with mean 1/2: $SD(\overline{Y}) = 0.5/\sqrt{n}$

> Coefficient analogue: $\sqrt{n}(\widehat{\beta} - \beta) \xrightarrow{d} N(0, \Omega)$

Measurement error creates a different source of inconsistency

A noisy regressor is not just a noisier outcome.

$$W = X + M, \quad Y = a + \beta W + (u - \beta M)$$

$$\operatorname{plim} \hat{b}_W = \beta \frac{\operatorname{Var}(X)}{\operatorname{Var}(X) + \operatorname{Var}(M)}$$

Check attenuation in a separate generated comparison

Use a clean outcome equation and known noise variance.

```
. regress clean_y noisy_x, vce(robust)

------------------------------------------------------------------------------
             |               Robust
     clean_y | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
     noisy_x |   .5177554   .0112879    45.87   0.000     .4956249    .5398859
       _cons |   1.045476   .0221565    47.19   0.000     1.002037    1.088915
------------------------------------------------------------------------------
```

Measurement error

Try the comparison before opening the explanation.

Practice

Observed x equals true X^* plus mean-zero noise independent of X^* and the outcome error. In a simple regression, what happens to the slope toward a positive true effect?

Measurement error: explanation

Follow the calculation and interpret its units.

> Under classical measurement-error assumptions, the probability limit is $\beta \frac{\mathrm{Var}(X^*)}{\mathrm{Var}(X^*) + \mathrm{Var}(noise)}$, which is between zero and β. More observations do not remove this attenuation. The result does not automatically extend to nonclassical error or every multiple-regression setting.

Outcome noise has another role

Mean-zero independent outcome error need not attenuate the slope.

> If the outcome is measured with an additional error unrelated to the regressors, that error enters the regression disturbance.

> The direction of measurement matters.

Missing observations can change the population represented

Complete-case analysis needs a selection argument.

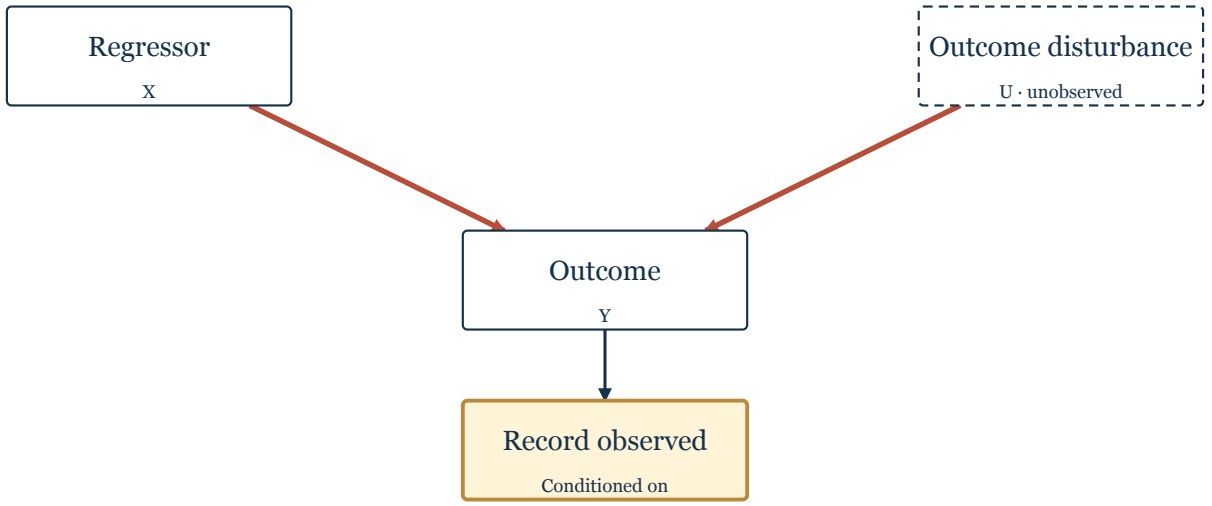


1 open directed route · 0 open alternatives · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

A missing-data assumption should be stated in context

Explain what makes the analysed cases informative.



0 open directed routes · 1 open alternative · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

Leverage and influence are different

An unusual regressor value need not be a bad observation.

> Leverage: unusual regressor values.

> Large residual: unusual outcome given the fitted predictors.

> Influence: material change in the fitted result when the case changes.

Influence

Try the comparison before opening the explanation.

Practice

A high-income district sits far from the remaining x values. Must it be deleted?

Influence: explanation

Follow the calculation and interpret its units.

> No. A distant x value can have high leverage; influence also depends on its residual and its effect on the fit. Verify the observation, explain substantive relevance and show defensible sensitivity analysis. Delete a confirmed error for a documented reason, not because it changes significance.

Sensitivity analysis answers a stated question

Change one defensible feature and explain the comparison.

> Sensitivity analysis examines how conclusions change under plausible alternative choices, such as a sample restriction, transformation or influential observation.

> For example, compare a fitted income relationship with and without one verified extreme-income record and show the range supported by each sample.

The data audit belongs before the policy sentence

Units, ranges and recording rules affect interpretation.

> Check how variables were measured, whether their ranges are possible and whether units changed over time or across sources.

> Record sample restrictions and verify identifiers before merging data.

Choose a remedy for the named problem

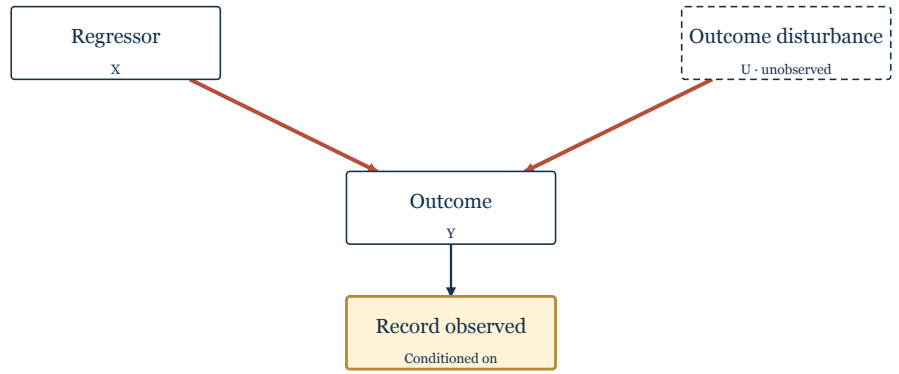
More observations, better measurement and better design do different jobs.

> If uncertainty is large under a credible comparison, more informative observations may help.

> State the problem before proposing a remedy.

DAG practice: selection on an outcome descendant

Trace the routes before choosing a control.



0 open directed routes · 1 open alternative · gold: conditioned on

Assumed causal structure · dashed boxes: unobserved · gold boxes: conditioned on

Practice

Suppose X and the outcome disturbance U are independent before selection, both affect Y, and a record is observed depending on Y. Why can restricting attention to observed records create an X–U association? Compare this with selection depending only on included X.

DAG practice: selection on an outcome descendant: explanation

Explain what conditioning changes in this stated graph.

> Y is a collider on $X \to Y \leftarrow U$. The observation indicator S is its descendant. Conditioning on $S = 1$ can open the collider path, so the selected sample can violate the disturbance mean assumption even when the original sample satisfied it. If selection instead depends only on included X and carries no extra information about U conditional on X, this particular path is not opened. The conditional slope may remain valid under a correct model and adequate support, although precision and the represented population can change. The selection mechanism, rather than missingness alone, determines the concern.

Separate simulation truth from estimated evidence

Reproduce the evidence, then explain its economic meaning.

Practice

Then use the actual schools.csv data. Fit score on staffing, language share and income; identify the largest Cook distance and compare the staffing slope with and without that district. Cook distance summarises how much omitting a case changes the fitted regression. Report this as sensitivity, not an instruction to delete it. Keep your commands, log, output and interpretation.

Separate simulation truth from estimated evidence: explanation

A reproducible result needs both a calculation and an interpretation.

> Simulation: direct coefficient 1; omitted-factor probability limit 1.5.

> Real-data sensitivity: staffing slope $-0.068776 \rightarrow -0.093391$.

> 420 versus 419 districts; investigate an influential case before deciding whether to exclude it.

Next, compare groups without misreading interactions

A group difference changes which coefficient combination matters.

> The simulation separated stable sampling behaviour from a valid target, while the data examples showed why measurement and selection need their own arguments.

> The next lesson introduces indicators and interactions.